Diary

2025-11-02

Tasks performed

- Trained a victim-detection model using CutMix as a data augmentation method.
- Drafted code for UnitV.

Issues & solutions

Issue	Solution
Observed that CutMix did not improve model accuracy at all.	Attempted to improve by using an attention-based data augmentation that can place donor images at appropriate locations.

Conclusion & future plan

- Since the target objects in the dataset are small, CutMix often causes information loss and likely did not improve accuracy (Fig.1). Kisantal et al. (2019) suggested that CutMix may not work effectively for small-object detection.[^1]
- As shown in Fig.2, the current model often fails to distinguish background from letter victims. Introducing attention-based CutMix is expected to produce a more robust and higher-accuracy model.

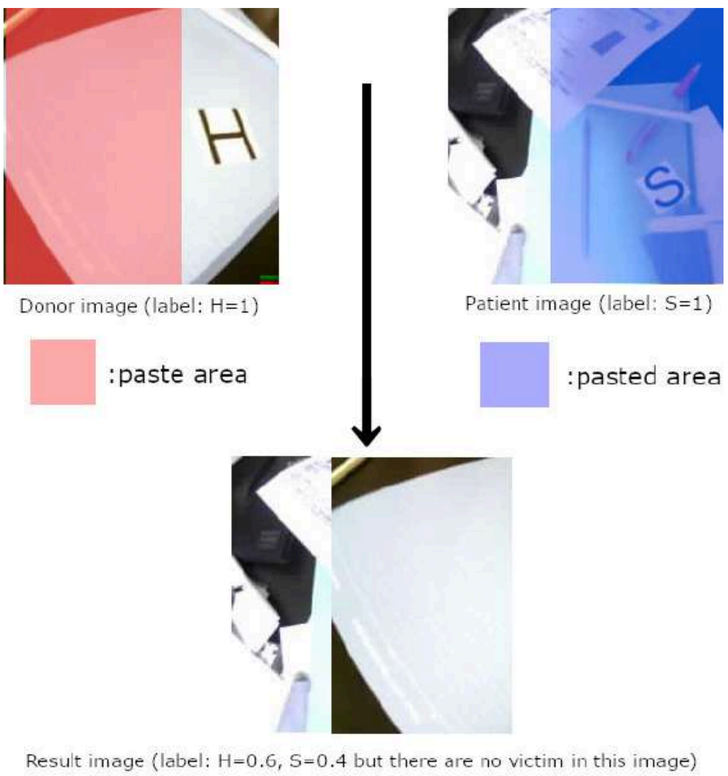


Fig.1 Information loss due to CutMix

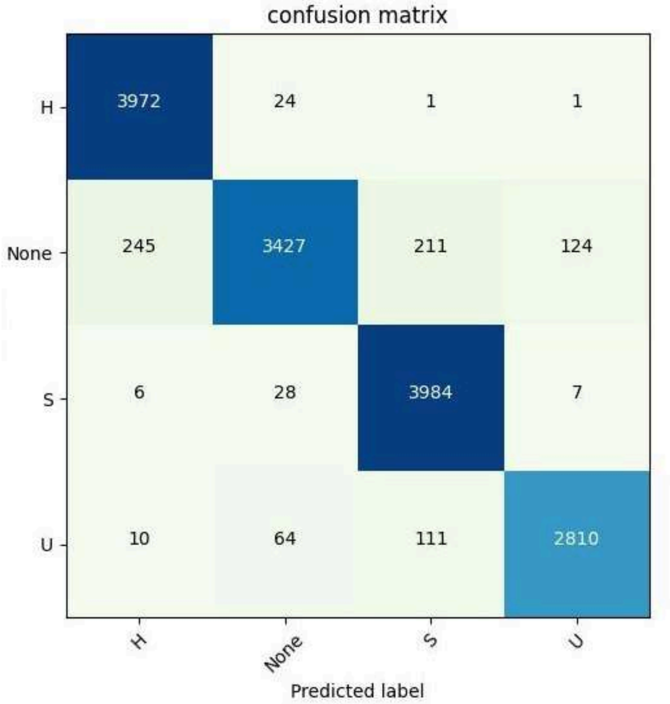


Fig.2 Confusion matrix of the current model

written by mitu

2025-11-03

Tasks performed

- Purchased electronic parts and mounted them on the PCBs.
- Verified operation of the STM microcontroller.
- Ordered 5268-3A from Marutsu; expected arrival 11/6 (Thu).

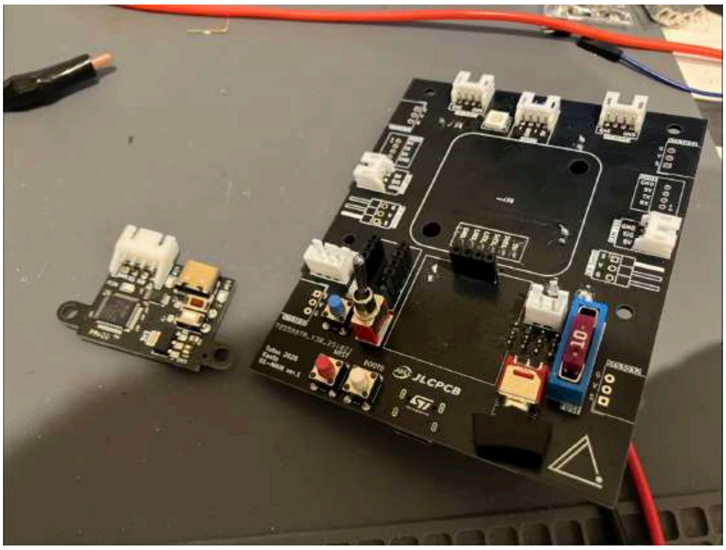
Issues & solutions

Issue	Solution
It was discovered that a PNP transistor was mistakenly used where an NPN transistor (Q6) was required in the low-voltage detection circuit on the 01-MAIN board.	Removed the PNP transistor and temporarily resolved by shorting the gate of Q3 (Pch MOSFET) to GND. Since the low-voltage detection is currently disabled, an NPN transistor will be installed once obtained.
A 4.7uF capacitor was forgotten on the VCAP pin of the STM32F446RE on the 01-MAIN board, allowing programming but preventing normal execution.	Recovered a 4.7uF capacitor from an older board and fixed the issue.
12V was mistakenly input to the 5V input on the 02-BOTTOM board, destroying a WS2812B that was powered by 5V.	Replace the WS2812B later.
L-shaped pin headers for the 01-MAIN board were out of stock.	Will purchase later.
The silkscreen for pin assignments on the back of the pin header for the rescue kit servo on the left side of the 01-MAIN board was incorrect.	Corrected with a pen.

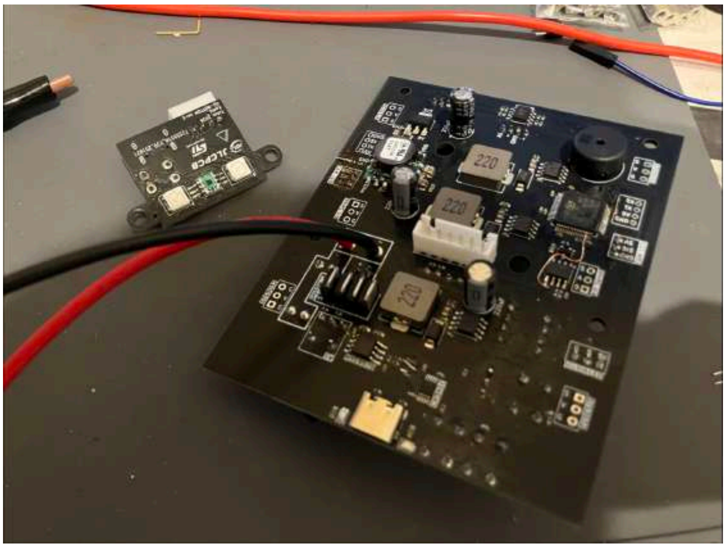
Conclusion & future plan

- Completed component mounting on the 01-MAIN board. The 5268-3A connector for STS3032, Grove connector for UnitV, and pin header for MG90S are not yet installed.

- Confirmed proper operation of the STM32 microcontroller, WS2812B, and buzzer on the 01-MAIN board.
- Completed all component mounting on the 02-MAIN board.
- Confirmed normal operation of the STM32 microcontroller on the 02-MAIN board. The WS2812B was likely destroyed by the 12V input and will be replaced later.
- Once 5268-3A connectors and L-shaped pin headers are obtained, they will be installed on the 01-MAIN board.
- Verify operation of the gyro sensor, display, and switches on the 01-MAIN board.
- Reinstall components on the 02-BOTTOM board and verify operation of WS2812B and S9706.



Board top (assembled)



Board bottom (assembled)

written by shuji

2025-11-05

Tasks performed

- Adopted an attention-based DA method for training.
- Changed the trainable layers from only the softmax layer to the whole model.
- Verified the trained model running on UnitV.

Issues & solutions

Issue	Solution
Very long generation time for attention caches during training.	Partially mitigated by saving caches in .npz format to avoid recomputing attention in subsequent runs.
Keeping attention at the original resolution (224x224) does not fit into memory.	Reduced memory by dividing the 224x224 image into 12x12=144 blocks and keeping only the pixel average within each block.
UnitV's SRAM constraints require the lightest firmware to use the kmodel, but that firmware lacks the `find_blobs()` function required to detect colored victims.	Unresolved. Search for custom firmware build tools on GitHub or consider implementing the function in C++ and flashing a custom build.

Conclusion & future plan

- As shown in a reference video[1], the initial aim of reducing false positives for letters H, S, U has been largely achieved. We consider letter-victim detection complete for now and will focus on communication programs and detecting colored victims.
- For colored victim detection, search for custom firmware build tools or implement a `find_blobs()` equivalent in C++.

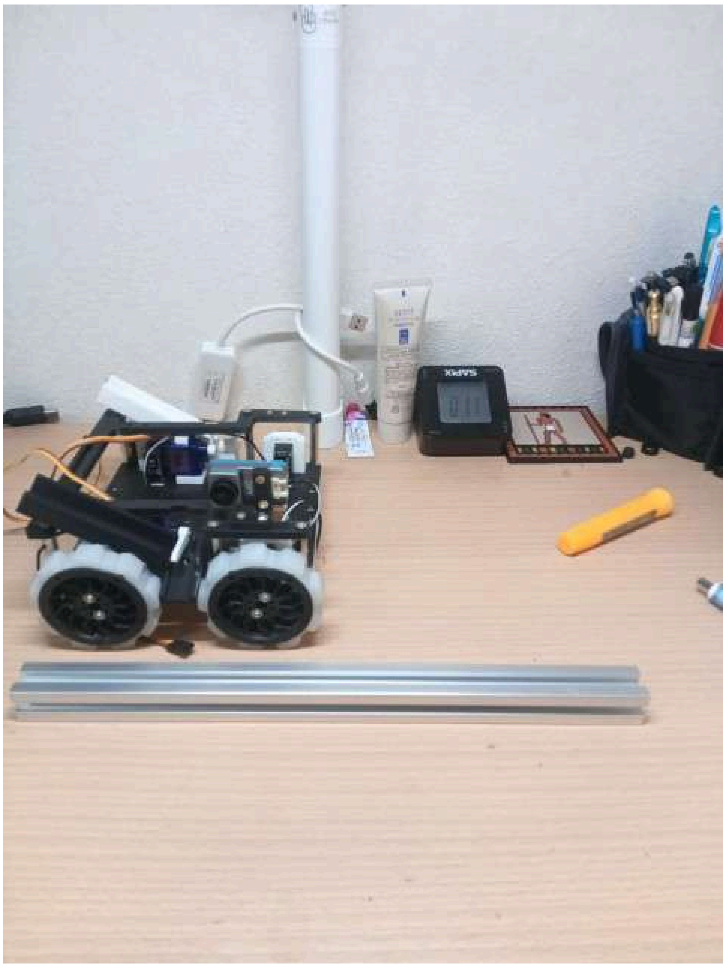
written by mitu

2025-11-06

Tasks performed

- Created three hard tires.
- Attached completed parts to the main body.
- Took photos.
- Added four more hard tires.

Current status


SAPIX


SAPIX
OPEN

Size comparison with small object

written by ryota

2025-11-06

Tasks performed

- Manufactured 3D-printed parts for the board unit.
- Performed a trial assembly of the board unit.

Conclusion & next plan

- Integrate the hardware and board unit on 11/07.

written by shuji

2025-11-07

Tasks performed

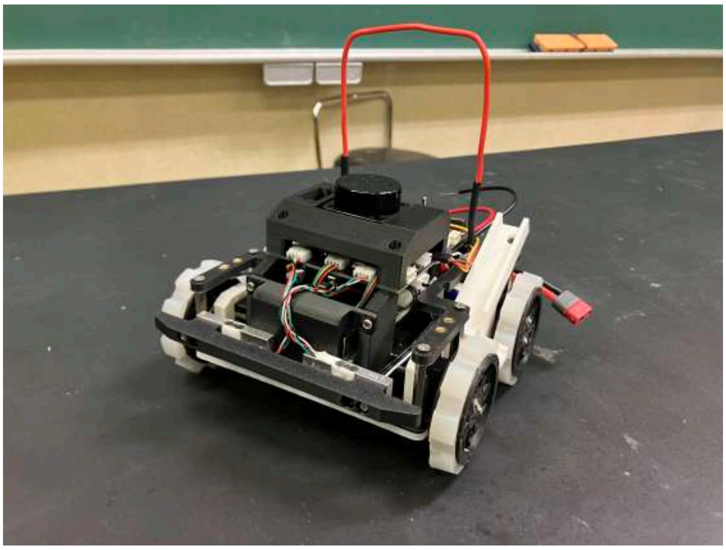
- Converted the load cell to use a PH connector.
- Created XH4 and XH6 pin cables.
- Integrated hardware and board unit (hardware assembly completed).
- Installed a voltmeter.
- Fixed board.

Issues & solutions

Issue	Solution
The cover and cover fastening screws of the 02-BOTTOM board interfere with holes in the bottom plate.	Reduce the thickness of the screw fastening area to address the interference.
The pin assignment of the 5268-03A connector for STS3032 on the 01-MAIN board was found to be reversed.	Implemented it on the back side to cope with it.
The voltmeter connector on the 01-MAIN board had reversed pin assignments compared with the handheld voltmeter.	Reinstalled the voltmeter connector on the 01-MAIN board in the opposite orientation.
Forgot the fan connector for Raspberry Pi.	Ran wires from the 5V/GND pins on the 04-RPi board.
Wires for servos etc. are long.	Shortening long wires is being considered because bundling them excessively causes noise.
Screw heads protruding under the bottom plate catch on a 20mm bump.	Use lower-profile screws.

Conclusion & next plan

- With the first machine's hardware tentatively completed, focus will shift to software development. The initial machine aims to be stable rather than highly capable, so the team will start software development early. Hardware team will concurrently start development of a second machine optimized for tougher terrain like Dangerous Zone.
- Sensor and motor checks will be performed on 11/08.
- Redesign the handle due to instability.
- Redesign the battery case.
- Consider wiring routing for servos.



Integrated hardware and board unit.

written by shuji

2025-11-08

Tasks performed

- Verified operation of STS3032: complete.
- Verified load cell: issues present.
- Verified SSD1306 display: complete.[2]
- Verified LiDAR from Raspberry Pi: issues present.
- Checked bump and slope tolerance.

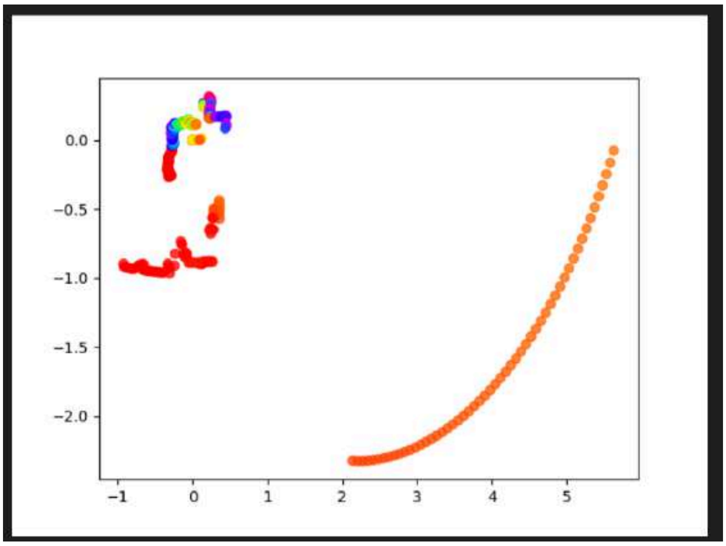
Issues & solutions

Issue	Solution
Discovered a footprint mistake for a PNP transistor in the RS485 converter circuit for STS3032.	Removed the component and replaced it with a spare 2SA1015.
Crimped wires came off the connector for the load cell.	Re-crimp on 11/10 (Mon).
Load cell values cannot be read.	Postpone investigation because it may be a connector-related issue.
High frequency communication errors with STS3032.	Temporarily mitigated by retransmitting data repeatedly; continue to seek root cause.
Frequent checksum errors when reading LiDAR from Raspberry Pi.	Found that an autostart program was running on the Raspberry Pi; disable it with systemctl.
LiDAR sometimes produces unnatural point-cloud maps.	Cause and countermeasure currently unknown.
The robot tips over on slopes greater than 25°.	Lower the center of gravity in the second machine to address this.

Conclusion & next plan

- Ran the robot and checked driving on slopes and bumps:
  - Can drive on slopes up to 25°.
  - Slightly catches on bumps ≤15mm but can still turn.
  - Can pass over bumps ≤25mm in a straight line.
- Continue to address unresolved issues below and verify BNO055 and rescue-kit servo operation which have not yet been checked:

- LiDAR anomalous data
- Load cell reading
- STS3032 communication reliability
- Decide software development responsibilities and software architecture soon.



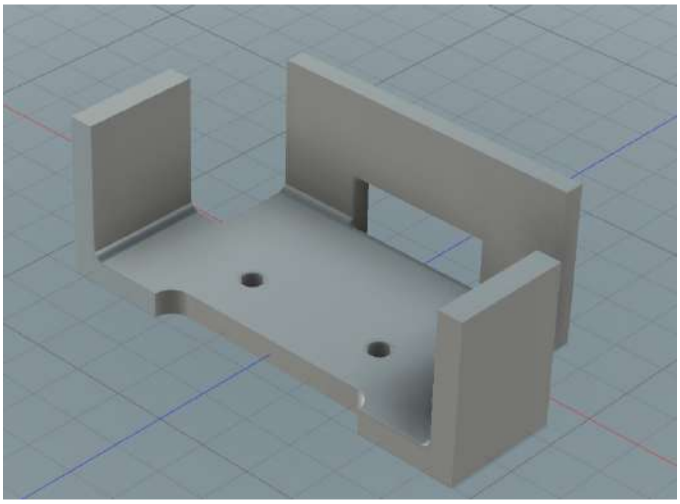
Anomalous LiDAR data

written by shuji

2025-11-10

Tasks performed

- Hardware design improvements
  - Battery case
    … changed the cushioning part relative to the board holder



Cutout of the affected area

- Rescue ejection mechanism
  temporarily fabricated a lid
  - Plastic plate for the arm
  - Modified the second-layer shape to fit the arm

written by ryouta

2025-11-11

Tasks performed

- Investigated LiDAR anomalies
- Verified BNO055 operation

Issues & solutions

Issue	Solution
Since the official LiDAR library worked without issues, the LiDAR hardware is fine and the custom LiDAR library was found to be faulty.	Rewrote the library.
BNO055 freezes soon after initialization.	Suspect I²C communication errors or a faulty BNO055; investigate by running it on another microcontroller or consider UART mode.
The load cell was wired to pins not supporting analog input (ADC_EXTI pins are ADC interrupt pins, not ADC input pins).	Fix in the next board revision.

Conclusion & next plan

- Rewrite the LiDAR library.
- Investigate the BNO055 malfunction.
- Complete verification this week and, based on this board's lessons, start design of the second machine's PCB after 11/17.

written by shuji

2025-11-12

Tasks performed

- Printed a handle
- Printed the second layer
  - Warped too much to be usable
  - Applied adhesive
  - Keep bed temperature

written by ryouta

2025-11-12

Tasks performed

- Fixed LiDAR problem
  - Rewrote the library.
- Built a test field
  - Created a field using aluminum wheels, M5 screws, and wooden boards.
- Field driving test[3]
  - Integrated the LiDAR-based wall detection with a preexisting exploration algorithm and succeeded in maze exploration.
  - Using constant control for straight and turn resulted in significant drift—intend to correct with LiDAR or gyro data.
  - This implementation was only to observe maze exploration; for the competition, all software will be rewritten. After hardware verification, prepare software specifications and begin development.

Conclusion & next plan

- The current robot was only driven to watch maze exploration, so software will be rewritten for the competition. After hardware verification, write proper software specifications and start development.
- Currently only able to create a roughly 3×3 field; will procure aluminum frames soon.

written by shuji

2025-11-14

Tasks performed

- Tested BNO055 with a different microcontroller
  - BNO055 itself worked fine.
- Verified rescue-kit servo operation

Issues & solutions

Issue	Solution
The microcontroller freezes when driving the servo.	Suspect noise on the servo power line; investigate further using an oscilloscope.
Suspected I2C bus issues.	Investigate by lowering pull-up resistors and other tests.

Conclusion & next plan

- Investigate servo noise.
- Investigate the I2C bus.

written by shuji

2025-11-15

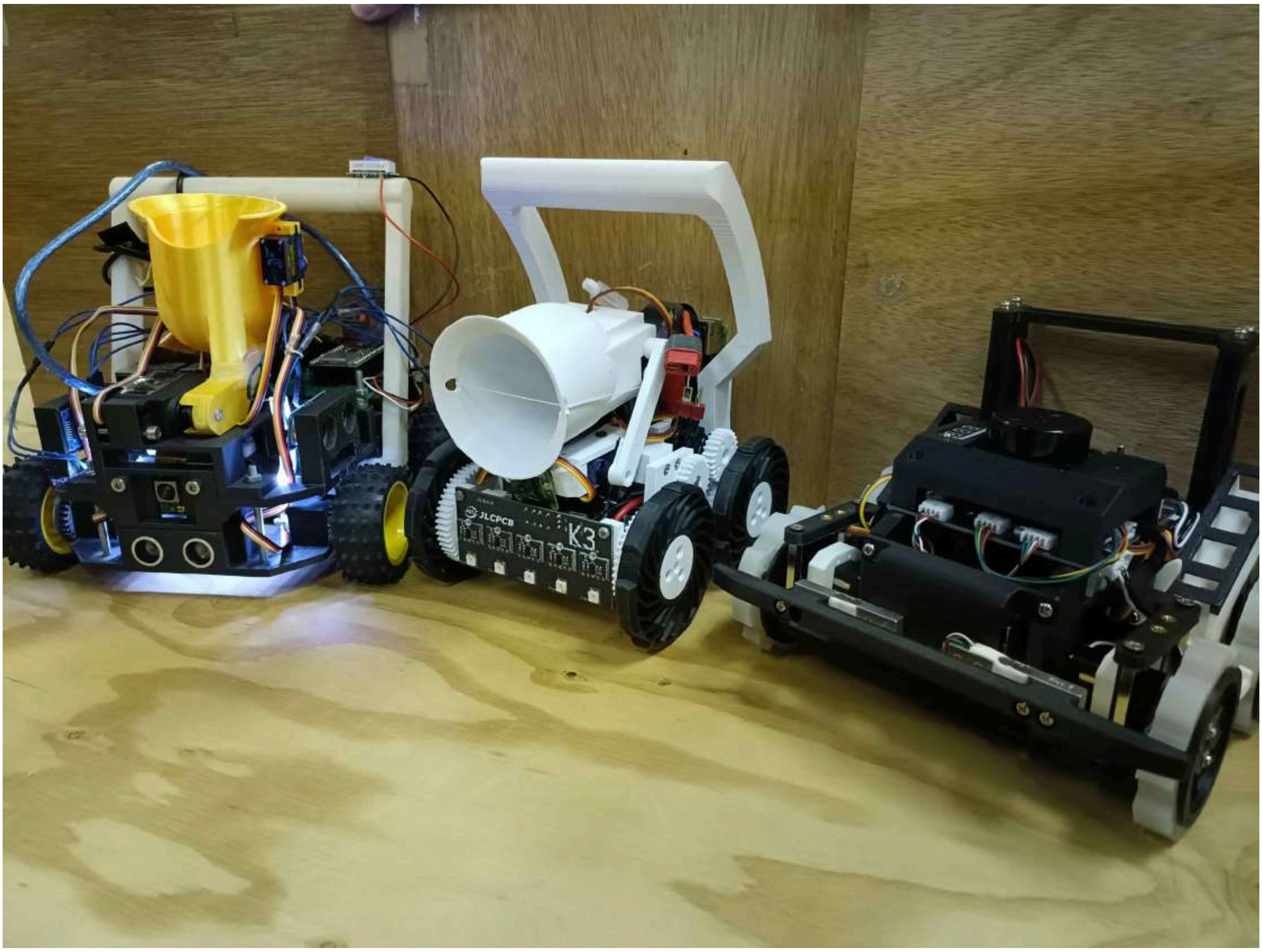
Tasks performed

- Replaced bottom plate
  - Redesigned the plate so screw heads do not protrude on the bottom.
  - Implemented the design on the real unit.
- Attached a lid to the rescue ejection mechanism.

Conclusion & next plan

- Bottom plate
  - Proceed without major changes for now.
- Rescue ejection mechanism
  - The current lid may snag the kit; change lid shape (e.g., rectangular cutout) to prevent snagging.
  - Investigate why the kit might be ejected at an angle and catch internally.
- Tires
  - The internal "k-shape" part of the tire catches on bumps. If using rubber like PTE, including a spring-like element inside the mechanism would be necessary and difficult.
  - Consider using rubber materials such as PTE to allow radial expansion/contraction for better handling of 20mm bumps.
  - Thin the outer ring of the tire to remove parts that catch on bumps.
- Battery case
  - Current case is worn and needs redesigning.
    - Plan to replace when reflecting the design that mounts a load cell at the rear.
  - Modify structure to allow mounting a load cell at the rear; adopt a hinge for the battery cover.

picture with junior



Tomorrow we supported a junior team at a local node competition and took photos.

written by ryouta

2025-11-16

Tasks performed

- Observed East Tokyo & West Tokyo node competitions
  - There was no Rescue Maze event, but observed Rescue Line and learned from it.
- Investigated servo issues
  - Under several conditions tested STM32F446RE and SG90 servo while observing power supply with an oscilloscope.
    - Normal supply[4]
      - Observed ripple about 1V with ~300Hz components.
      - Reset occurs after moving a servo once.
    - Using stabilized power supply[5]
      - Observed only small noise (~0.1–2V ripple).
      - MCU does not reset and servo continues to move.
      - Determined that servo power was the cause.
    - After adding a capacitor

- Adding a 4000uF electrolytic capacitor to the servo power reduced ripple to about 0.3V and prevented MCU resets.
- Large capacitor caused MCU not to boot on power-on; pressing reset after power-on boots it.
- Supplying the MCU 3.3V from another microcontroller (Nucleo F303K8) and driving the servo from Nucleo
  - Operated without problems.
  - Determined noise is not entering via MCU power or GPIO.
- Reinstalled previously added 47uF VCAP for the MCU.[6]
  - The problem was resolved.
  - The VCAP being located above the power line and being poorly hand-soldered likely caused instability due to noise.

Conclusion & next plan

- Solved the servo problem. For the second machine, place components and route power so actuator noise does not affect control or sensors.

written by shuji

2025-11-17

Tasks performed

- Investigated I2C (BNO055 related)
- Meeting about software architecture
  - Considered which functions run on which processors
  - Divided software development responsibilities
- Load cell reading[7]
  - Rewired to ADC-capable pin and confirmed readings.

Issues & solutions

Issue	Solution
Lowering pull-up resistors improved I2C but stability remains an issue.	Consider board layout changes so I2C traces are as short as possible on the second machine and calculate appropriate pull-up values.
Mapping, path planning, and localization are hard to split; splitting responsibilities complicates development.	Do not split logic ownership; ensure software is readable and documented so either developer can continue.
Load cell readings are unstable when no load.	The amplification circuit uses inverting op-amp with very high gain, so accuracy suffers; consider HX711 ADC as alternative.
Heard high-frequency (12–16kHz) noise from STS3032, probably unavoidable and not operationally problematic.	Not solvable; no operational impact.

Conclusion & next plan

- STM handles minimal functions: send sensor values to Raspberry Pi and execute actuator commands from Raspberry Pi. Logic processed mainly on Raspberry Pi.
- All communication initiated from Raspberry Pi.
- STM uses interrupt-driven UART receive to respond quickly.
- Shuji mainly handles STM code and Raspberry Pi device classes; Mitu handles UnitV image processing and logic, but both should be able to develop any part with readable code and documentation.
- Raspberry Pi code will be written in Python with type hints, dataclasses, and Enums; document with Doxygen-style comments.
- Device-related class design is substantially complete; begin communication design.
- With hardware checks done, begin second-machine development and finalize hardware specs soon.

written by shuji

2025-11-18

Tasks performed

- Implemented and verified communication between STM32F446RE and Raspberry Pi.
- Wrote a blog post about the first machine "Kuro" PCB.

Issues & solutions

Issue	Solution
Servo movement during rescue kit drop caused communication to be unavailable during that period.	Implemented asynchronous handling by queuing servo actions with trigger times.
Blog index page navigation was not working.	Could not find root cause; increased items per page as a temporary workaround.

Conclusion & next plan

- Communication between STM32F446RE and Raspberry Pi is complete (some peripherals like buzzer and display remain unimplemented).
- Also implement communication between STM32F103C8 and Raspberry Pi.

written by shuji

2025-11-19

Tasks performed

- Created a provisional software documentation site hosted with Sphinx at https://rcj2026-documentations.onrender.com/.
- Advanced implementation of the algorithm on Raspberry Pi to a basic level excluding slopes, obstacle avoidance, and black-tile escape.

Issues & solutions

Issue	Solution
import statements did not behave as expected	Consulted official Python documentation to correct misunderstandings about imports.
.md rendering with Sphinx differed from .rst appearance	Introduced myst_parser to allow mixing rst directives in markdown and fixed appearance.

Conclusion & next plan

- Continue enriching software documentation.
- Implement angle estimation using LiDAR since gyro sensor communication is unavailable.
- Implement slope, obstacle avoidance, and black-tile escape.
- Test and debug on the real robot.

written by mitu

2025-11-19

Tasks performed

- Verified load cell readings using HX711.
- Decided connector and part placement for the second machine.
- Completed the main PCB schematic for the second machine.

Issues & solutions

Issue	Solution
Instability when no load remained even with HX711.	Determined HX711 did not improve instability; will continue using op-amp inverting amplifier for load cell readings.

Conclusion & next plan

- Considering switching bumper sensors to micro-switches due to stability and durability concerns; include connectors for both load cell and micro-switch on the board.

written by shuji

2025-11-20

Tasks performed

- Continued circuit design for the second machine.

Conclusion & next plan

- Decided to switch gyro sensor to BNO085 for more stable operation; plan to purchase from Digi-Key.

written by shuji

2025-11-21

Tasks performed

- Improved BNO055 operation.

Issues & solutions

Issue	Solution
BNO055 used thin pin headers and unstable connection with regular sockets was suspected cause of freezes.	Soldered directly to the PCB instead of using sockets.
Freezes were reduced but still occasional.	Implemented I2C bus recovery on freeze to continue measurement.
Angle sometimes jumps by ~180 degrees while rotating.	Unknown cause; possibly a BNO mode issue; further investigation required.

Conclusion & next plan

- BNO055 operation improved to a usable level, but angle jump issue unresolved.
- Continue investigating; still plan to adopt BNO085.

written by shuji

2025-11-22

Tasks performed

- Purchased parts at Akizuki Denshi.

written by shuji

2025-11-23

Tasks performed

- Created a prototype micro-switch bumper.[8]

Conclusion

- AV620264 micro-switch can be used to build a practical bumper.

written by shuji

2025-12-01

Tasks performed

- PCB design completed and ordered.

Conclusion & next plan

- PCBs expected around 12/11; prepare parts and assemble upon arrival.

written by shuji

2025-12-09

Tasks performed

- Parts procurement
- Ordered PCBs arrived; started assembly.

Issues & solutions

Issue	Solution
DCDC on 01-MAIN board makes abnormal noise when powered.	Investigated these but did not improve: replaced power cable, fuse and holder, removed reverse-protection circuit, re-soldered DCDC.

Conclusion & next plan

- Suspected causes:
  - Low-voltage detection circuit error
  - Poor soldering/implementation of OKL-T/6-W12N-C
- Plan: test power protection circuits on a breadboard and remove OKL-T/6-W12N-C to probe with an oscilloscope.
- Resume development after final exams.

written by shuji

2025-12-10

Tasks performed

- Investigated malfunction of 01-MAIN board

- Observed behavior with regulated power supply.
    - Board operates normally when supplied with ≳12.0V. When lowering voltage, low-voltage detection triggered around 10.5V and shut power off.
    - Abnormal noise occurs when supply is below ~12.0V and persists even when raised to ~12.6V.
    - Applying voltage directly to the output of the power cutoff circuit did not reproduce the abnormality at any voltage.
    - Concluded instability around the low-voltage detection threshold is the cause. Using the voltage-detection IC output directly to drive a MOSFET for cutoff may have caused instability.
    - Disabled the voltage detection IC since its importance is low.
- Verified BNO085 by soldering headers directly for reliability.

Issue	Solution
CH340E serial converter not recognized when 01-MAIN board connected to PC via USB.	After implementing power management components, the two affected boards showed the issue while untouched boards did not; concluded CH340E may have been damaged by prior faults. Re-implement on a new board to resolve.

Conclusion & next plan

- 01-MAIN board assembly mostly complete.
- Install remaining components on 03-ToF and 04-RPi boards.
- Aim to finish the new machine by 12/12 and shift to software development.

written by shuji

2025-12-11

Tasks performed

- Completed soldering.

Issues & solutions

Issue	Solution
Variation in measurement range among VL53L0X units.	Selected the four with the least variation to implement.

Conclusion & next plan

- Continue design and manufacturing of 3D-print parts for board units.

written by shuji

2025-12-14

Tasks performed

- Switched to TPE tires.

Issues & solutions

- Because everything was made in TPE, tire shape deformed and driving became difficult.
  → Plan to attach a metal circular part to the outer circumference to restore shape.

written by ryouta

2025-12-15

Tasks performed

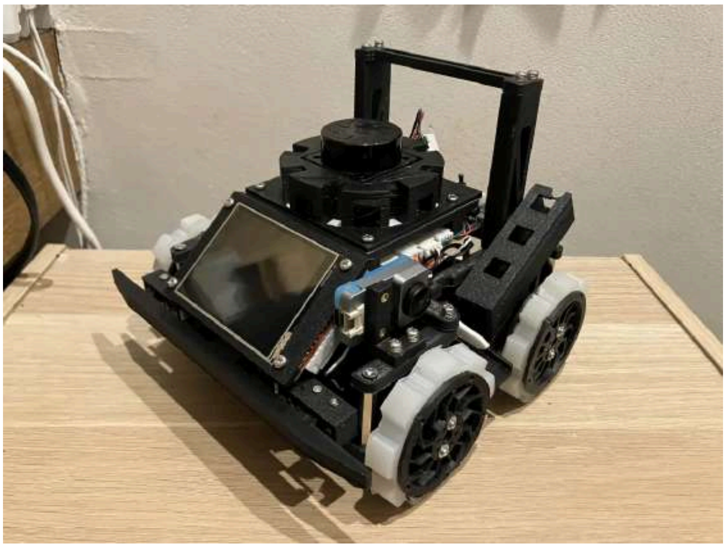
- Improved 3D-printed parts for board units.
- Completed and installed micro-switch bumper.
- Improved firmware for ToF board.

Issues & solutions

Issue	Solution
Used `readRangeContinuousMillimeters()` from Pololu VL53L0X library which blocks until measurement completes, stalling the main loop and affecting communication with main board.	Monitor internal register `0x13` bit that indicates measurement completion and call `readRangeContinuousMillimeters()` only when ready for non-blocking behavior, reducing main-loop time from ~50ms to ~3ms.

Conclusion & next plan

- Continue software development.



Current robot with bumper and display attached.

written by shuji

2025-12-16

Tasks performed

- Modified cover for the 02-BOTTOM board (color sensor) and flashed programs to the rear board.
- Enabled SPI on Raspberry Pi and verified display operation.

Issues & solutions

Issue	Solution
`/dev/spidev0.0` not created even after enabling SPI.	UART4 was occupying pins. Disabled UART4 in `/boot/firmware/config.txt` to resolve.
Display not working.	Unresolved.

Conclusion & next plan

- Continue verifying display operation.

written by shuji

2025-12-18

Tasks performed

- Performed basic driving tests with the second machine.[9]

Issues & solutions

Issue	Solution
Communication delay with STM during rotation causes deviation.	Switch to coarse fast rotation followed by slower fine adjustments.
Collisions with pillars lead to LoP.	Rotate a small amount opposite the bumper direction then move forward; also improves obstacle avoidance.

Conclusion & next plan

- Basic driving tests completed.
- Continue work on black/blue/silver tiles, slopes, and rescue kit handling.

written by mitu

2025-12-19

Tasks performed

- Implemented black-tile handling.[10]
- Verified color sensor operation.
- Verified new ToF sensors on second machine.
- Checked slope and step resilience.[11][12]
- Tested UnitV on actual field.

Issues & solutions

Issue	Solution
Rear color sensor not working.	Unresolved; low-level hardware team investigating.
On slopes without sufficient wall height at the top, cannot stop at tile center.	Consider multiplying travel distance by cos(theta) using gyro to correct movement.
STM communication not functioning properly.	The expected byte count from STM was incorrect; fixed by changing to the correct count.
Gyro sensor values not updating correctly.	STM was returning a pointer rather than a value; fixed to return value and resolved the issue.
UnitV's current field-of-view may be insufficient to reliably detect victims.	Move UnitV mounting from robot edges to above LiDAR; hardware team is designing this.
UnitV fails when victim is too close.	Add images of victims at closer distances to dataset to improve detection.

Conclusion & next plan

- Continue software development for blue/silver tiles, slopes, and rescue kit handling.

written by mitu

2025-12-20

Tasks performed

- Augmented dataset for close-range victim detection.
- Trained new models on the augmented dataset.
- Deployed trained models to UnitV and tested operation.[13]
- Implemented placeholder implementations for slopes and tiles without hardware.

Issues & solutions

Issue	Solution
Accuracy for U victims not as high as expected.	Expand dataset and retrain. Also handle cases where U is fully in-frame by software heuristics.

Conclusion & next plan

- Perform field tests of the provisional slope/blue/silver tile implementations after SSH presentation at Kogakuin University.

written by mitu

2025-12-22

Tasks performed

- Field tests for blue and silver tiles.[14][15]
- Slope field tests (no video).

Issues & solutions

Issue	Solution
Gyro sensor stops returning valid values during driving.	Sent int16_t signed values as-is from STM to RPi, which overflowed; changed RPi to receive as unsigned to handle the wrapping.
Rotation direction incorrect sometimes.	Normalized angles during rotation to prevent errors near 0°.
Unable to detect downhill.	Changed detection to use `min(current_angle, 360-current_angle)` to also catch downhill angles (around 340°).

Conclusion & next plan

- Blue/silver tile and slope handling are mostly completed.
- Continue rescue-kit work next.

written by mitu

2025-12-24

Tasks performed

- Assembled spare machine PCB.[16]

Issues & solutions

Issue point	Solution
UART between 01-MAIN board and camera was unreliable.	Reflowed solder joints to fix.

written by shuji

2025-12-28

Tasks performed

- Driving tests.

Issues & solutions

Issue	Solution
Some walls in the course are occasionally misdetected as yellow victims.	Considered not a major issue since competition walls are expected to be white; consider adjusting UnitV exposure.
Victim detection fails when camera is too close to the wall.	Unresolved. Consider changing UnitV mounting angle, double-checking walls, or keeping distance with control changes.
Cannot distinguish black tile and bump.	Add front Raspberry Pi camera to assist tile classification.

Conclusion & next plan

- Continue software development and fix bugs as found.
- Use Raspberry Pi camera to assist tile classification.

written by mitu

2025-12-29

- Driving tests performed.
- Successfully dropped the rescue kit for the first time.[17]

Issues & solutions

Issue	Solution
Running over origami silver tile can short-circuit in some cases.	Cause unknown; suspected contact between origami and metal parts. Using aluminum sheet for silver tiles eliminated the issue.
Two rescue kits can stack vertically when dropped.	Offset drop angle by ~15° to avoid stacking.
Servo sticks during rescue kit drop.	Designing improved mechanism.
Rescue kit stepping on causes robot to move >15cm away from victim sometimes.	Unresolved; consider asymmetric-weight kits.

Conclusion & next plan

- Verify improved rescue-kit drop mechanism.
- Continue software development.

written by mitu

2026-01-01

Tasks performed

- Driving tests.
- Verified improved rescue-kit drop mechanism.
- Investigated and fixed UnitV `sensor.set_windowing(x,y,w,h)` where x,y were ignored.
- Investigated and fixed `sensor.set_auto_exposure(False, exposure_us=...)` manual exposure not taking effect.
- Investigated and fixed `sensor.set_auto_gain(False, gain_db=...)` manual gain always zero issue.
- Verified firmware operation on actual hardware.

Issues & solutions

Issue	Solution
Current color sensor cannot distinguish silver and white tiles.	Plan to install photo-reflector under the chassis.
`sensor.set_windowing` : width/height change but x/y ignored (position fixed).	Common firmware code passed only w,h to the driver. Implemented sensor internal crop in OV7740 driver and pass x,y,w,h to OV7740-specific driver while fallback drivers still receive w,h only.
`sensor.set_auto_exposure(False, exposure_us=...)` ineffective.	The argument was not in microseconds but in exposure line units; convert `exposure_us` to line counts using tROW (33us) for OV7740.
`sensor.set_auto_gain(False, gain_db=...)` becomes zero every time.	`set_gainceiling` was overwriting the register entirely; update only relevant bits to preserve gain_db.

Conclusion & next plan

- Confirmed ROI, manual exposure, and manual gain work on UnitV.
- Continue development of photo-reflector-based silver tile detection.
- Continue driving tests.

written by mitu

2026-01-12 Kanto Block Competition

Tasks performed

- Participated in the Kanto Block competition.

Issues found and fixes during competition

Items fixed during the event

Issue	Fix
When going up stairs forward/back walls disappear and straight motion cannot continue.	Apply same implementation used for slope tiles where control was set to fixed-speed.

Issue	Fix
DangerousZone misdetected as a red victim.	Fixed a bug related to UnitV latency that caused distant objects behind walls to be misdetected.

Remaining issues

Issue	Fix
High center of gravity causes tip-over on 25° slope with a 20mm bump.	Review component placement and make metal tires.
Rarely gets stuck on 20mm round bumps.	Increase tire diameter.
VL53L0X ToF long-range accuracy poor, unusable.	Consider alternative ToF sensors.
Hard to distinguish ramp vs wall.	Mount ToF sensors vertically separated to distinguish by distance differential.
Buzzer volume too low.	Increase volume.
No display.	Add display or equivalent to show current position etc.
Rear bumper activated from a push on the opposite side.	Redesign bumper so it only reacts to frontal pushes.
Digit 8 misdetected as S.	Augment training data with digits and other symbols.

Conclusion & next plan

- Participated and achieved competition victory and poster award.
- Details on the competition are on the blog: https://tuton-rcj.github.io/20260114/

2026-01-15

Tasks performed

- Verified operation of VL53L4CX.
- Designed ToF module using VL53L4CX.

Conclusion & next plan

- VL53L4CX provides stable measurements within 180cm with <10mm error.
- Beyond 180cm, RangeStatus becomes `4 : out of bounds` but measurements up to ~250cm are still usable.
- Accurate measurements beyond 250cm are difficult under normal lighting.

VL53L4CX test

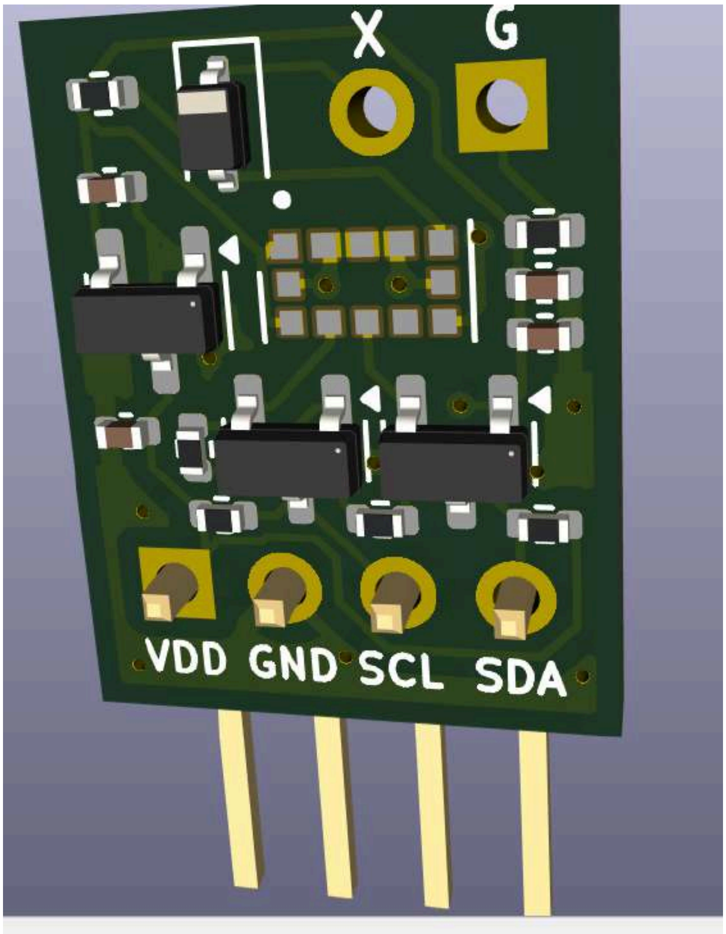
- Chose VL53L4CX for the national competition machine and started ToF module schematic design; board completion planned by 01/16.
- Consider adding extra microcontrollers by exposing UART from Raspberry Pi USB port.

written by shuji

2026-01-16

Tasks performed

- Completed and ordered PCB design for ToF module using VL53L4CX.



ToF module

Conclusion & next plan

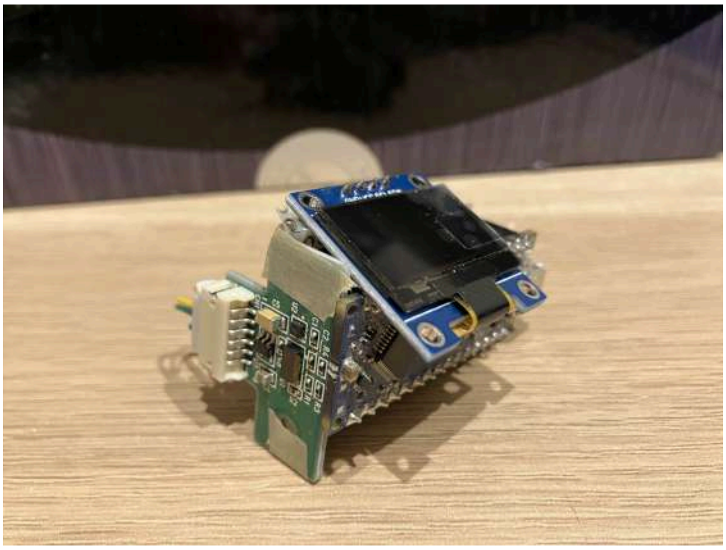
- ToF module boards expected around 1/24.

written by shuji

2026-01-19

Tasks performed

- Created a ToF & display module using Arduino Nano Every connected to Raspberry Pi's USB.
- Combined Arduino Nano Every, Akizuki VL53L0X module, and OLED display.



ToF & display module

Conclusion & next plan

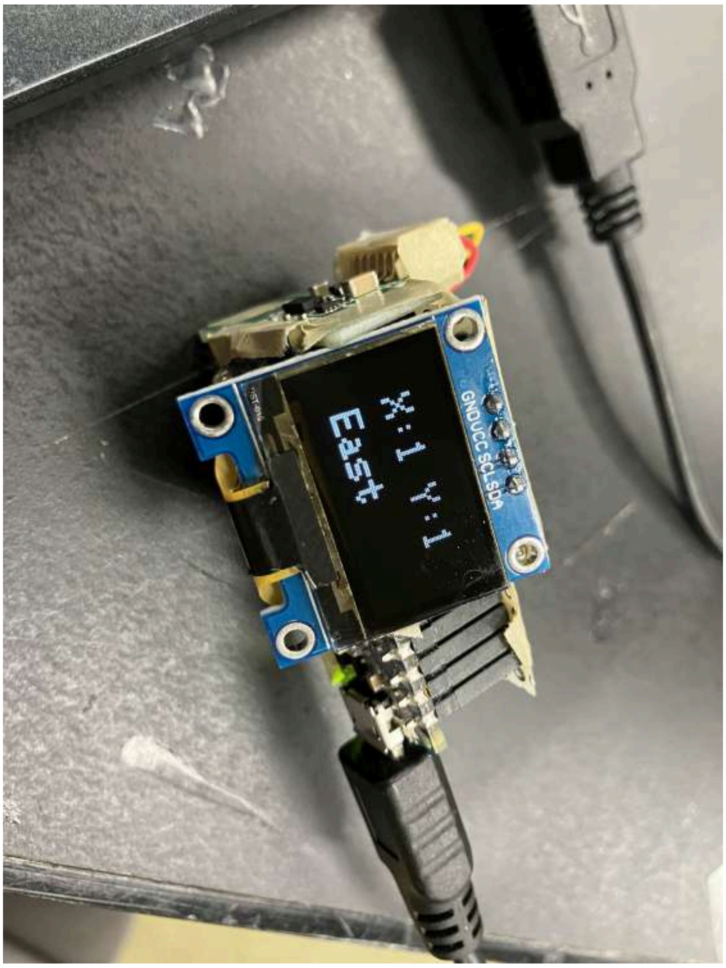
- Write firmware for this module.

written by shuji

2026-01-20

Tasks performed

- Wrote firmware for the ToF & display module.
- Implemented function to respond to ToF requests from Raspberry Pi.
- Implemented display of coordinates and orientation sent from Raspberry Pi to OLED.



Display showing coordinates

Conclusion & next plan

- Implement Raspberry Pi side communication with the module.

written by shuji

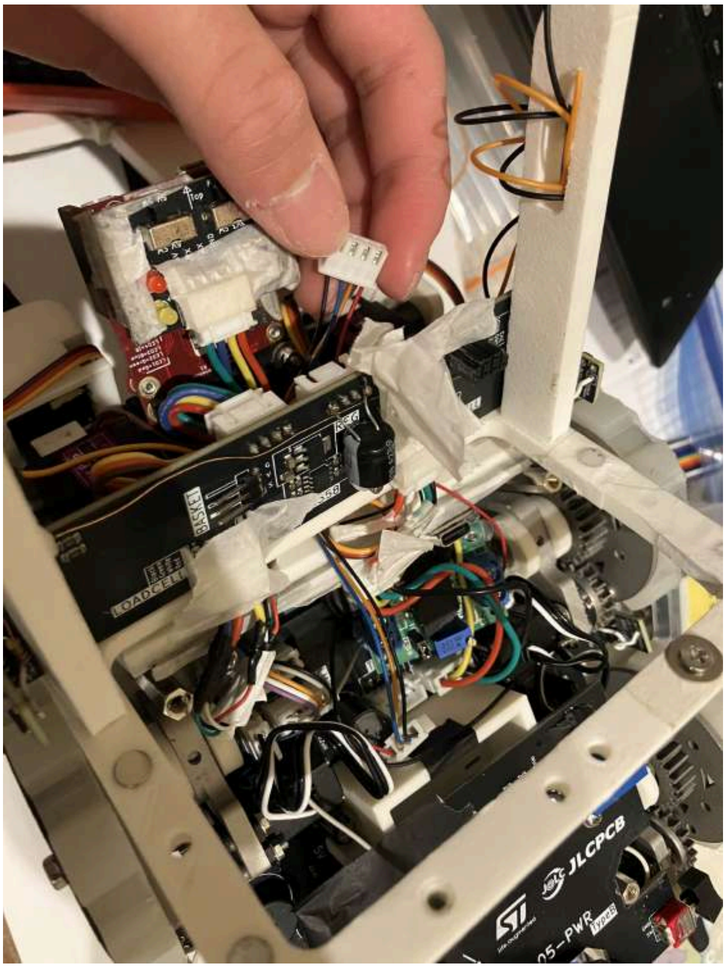
2026-01-21

Tasks performed

- Serviced the 2025-season Rescue Line national machine "Shiro".
- Participated in Student Robot EXPO and prepared the Rescue Line machine since upcoming entrance exams may reduce available development time next season.

Issues & solutions

Issue	Solution
Cable between layer 2 and layer 1 is broken.	Create a new cable.



Inside Shiro

Conclusion & next plan

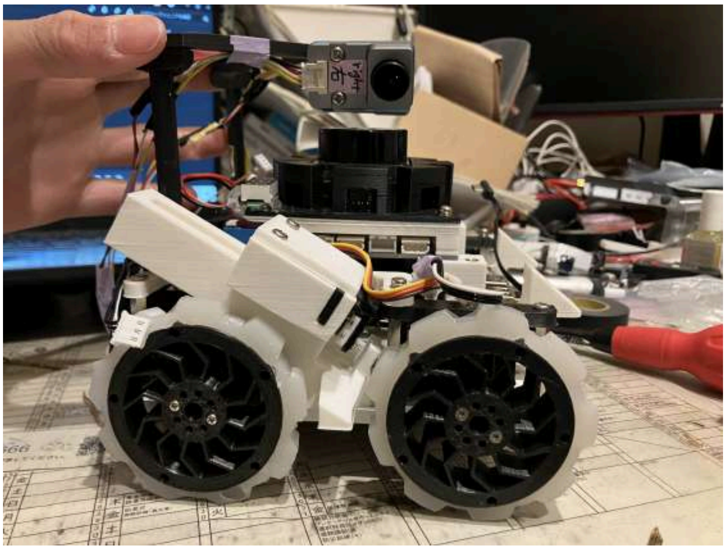
- Finish cable and verify Shiro operation.

written by shuji

2026-01-22〜2026-01-24

Tasks performed

- Implemented VL53L4CX ToF module on 03-ToF board.
- Reviewed orientation and placement of Raspberry Pi, 01-MAIN, and 03-ToF boards and lowered overall height by ~5mm.



The LiDAR used to be close to UnitV camera; overall board modules are lowered.

Issues & solutions

Issue	Solution
6-pin XH connector connecting 01-MAIN and 04-RPi interferes.	Moved the XH connector on 04-RPi to avoid interference.
Capacitor on 01-MAIN interferes with 04-RPi board.	Solder the capacitor sideways.
40-pin socket on 04-RPi is hard to plug/unplug.	Use Raspberry Pi specific 40-pin socket.

Conclusion & next plan

- Adjust components affected by hardware refinements.

written by shuji

2026-01-27

Tasks performed

- Verified Shiro operation; completed cable creation.[18]
- Adopted knurled screws for battery cover.[19]

Conclusion & next plan

- Continue tuning Shiro using the field.

written by shuji

2026-01-28

Tasks performed

- ToF software development and verification.

ToF verification

- Started writing TDP for the world championship.

Issues & solutions

Issue	Solution
ToF data communication takes ~9ms per sensor which is slow.	Increase I2C clock to 800kHz.
ToF measurement time long.	Reduce ToF budget time from 33ms to 15ms; sufficient for ~2m range.
Reading all four sensors still takes ~6ms even at higher speed; request-response from 01-MAIN to 03-ToF may introduce latency.	Change to push method: 03-ToF board sends data automatically when measurement is ready.
03-ToF and 01-MAIN board connection unstable.	Consider reflowing connector soldering.

Conclusion & next plan

- ToF software development completed.
- Assemble hardware and aim to start software development after the early-February holiday.

written by shuji

2026-01-30

Tasks performed

- Created speaker module for Raspberry Pi to play audio from a 3.5mm stereo jack.
  - Video: [20]

written by shuji

2026-02-02

Tasks performed

- 3D-printed camera mount parts
- Assembled robot
- Built and tested a broom prototype for debris removal

Conclusion & next plan

- Assembly except for tires, handles, camera LED, and photo-reflector equipped color sensor is complete.
- The broom can remove debris like toothpicks and tissues, but very flat paper remains and slides off slopes.
- No interference between broom and tires.
- Start software development.

written by shuji

2026-02-03

Tasks performed

- Started building practice field for national/world competitions.
- Purchased two 910×1820×9mm wooden boards and drilled holes for posts at 30cm intervals.
- Used 20×20×200mm aluminum frames as posts screwed into the boards.

Conclusion & next plan

Field construction

- Make walls from corrugated plastic tomorrow.

written by shuji

2026-02-04

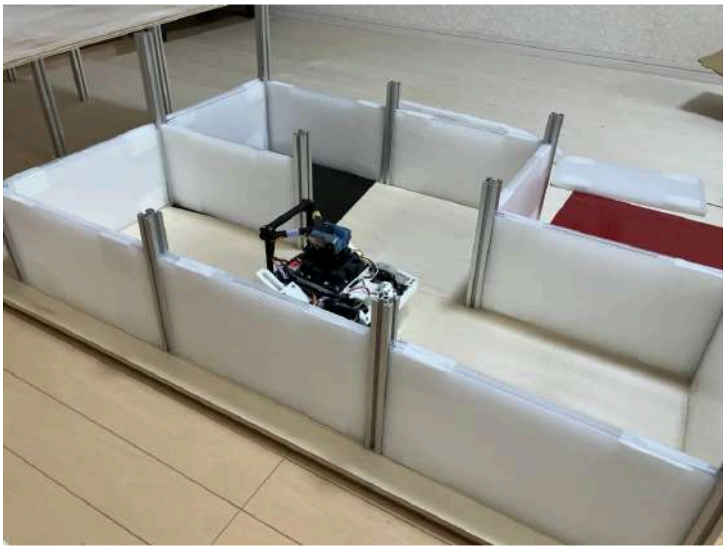
Tasks performed

- Built field walls using 910×1820×2.5mm protective panels.
- Created 150mm-high walls that slide into the aluminum frames; one panel yields six walls.

Protective panel

Walls

Conclusion & next plan



Field

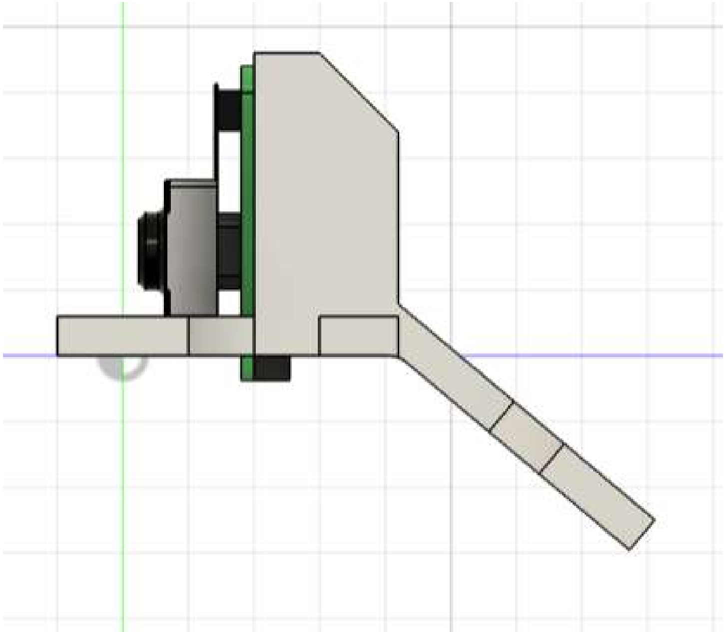
- Need about 60 walls total; continue making them (12 made so far).

written by shuji

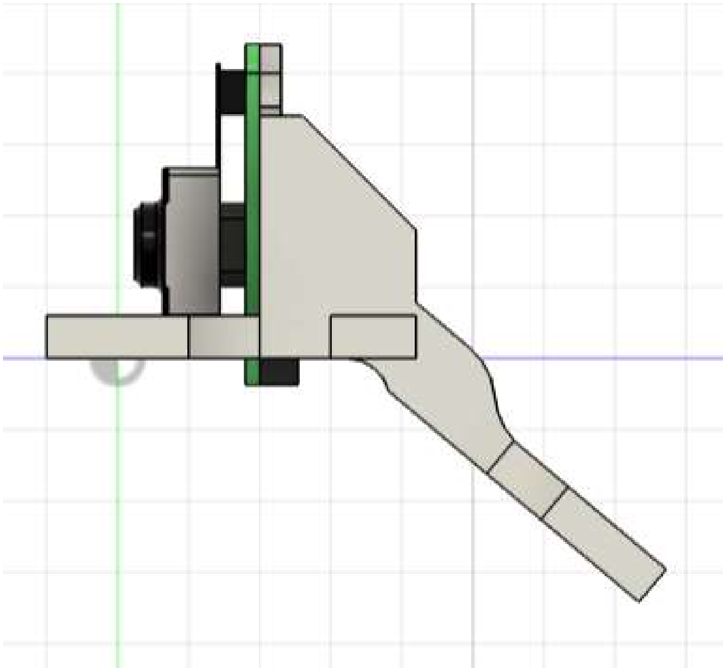
2026-02-07

Tasks performed

- Improved camera mount parts after lowering ToF sensor position to avoid slight interference.



before



after

written by shuji

2026-02-09

Tasks performed

- Mounted LEDs on the top of the robot for camera illumination and victim-found indicator.
  - Video: [21]

Conclusion & next plan

- LED mounting completed on the spare machine; install on main machine and use LEDs to indicate victim detection.

written by shuji

2026-02-11

Tasks performed

- Added a Raspberry Pi 4 for the second machine (previously used Pi 5).
- Improved bumper design:
  - Previous bumper had connected left/right sides causing opposite-side sensor triggers when pushed from the rear.
  - Made left/right bumpers independent to fix this.
- Added two photo-reflectors to the color sensor.
- Changed Shiro's camera from OpenMV to UnitV and designed/3D-printed a UnitV mount.

Issues & solutions

Issue	Solution
libcamera library not usable.	Specify `--system-site-packages` when creating the venv.

Conclusion & next plan

- Raspberry Pi initial setup documented: https://hackmd.io/@shuji4649/r1xOuLtDZx

written by shuji

2026-02-14

Tasks performed

- Fixed connectivity issue where second machine's Raspberry Pi wouldn't connect to mobile hotspot.
- Confirmed second machine operating normally.

Issues & solutions

Issue	Solution
Second machine Raspberry Pi disconnected from mobile hotspot.	Reconfigured Wi-Fi.
Servo IDs differed between machines.	Adjusted IDs to match.

Conclusion & next plan

- With the second machine working, use it to accelerate software development.

written by shuji

2026-02-22,23

Tasks performed

- Participated in Student Robot EXPO and exchanged ideas with many teams.

Issues & solutions

Issue	Solution
LiDAR point cloud abnormal at venue.	Infrared sources in the venue likely interfered with LiDAR.

Conclusion & next plan

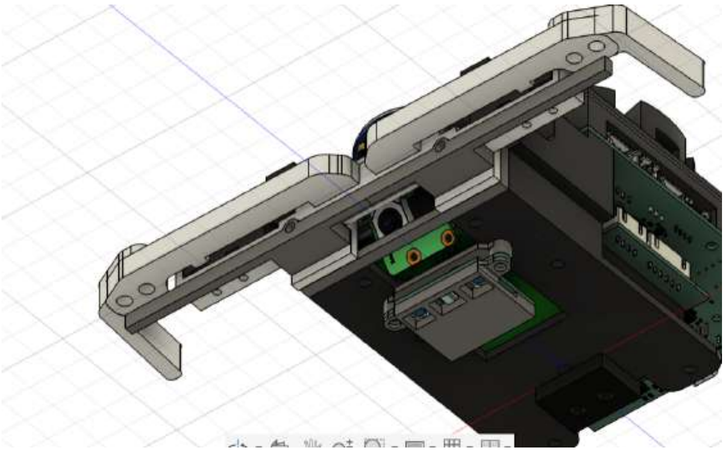
- Prepare backup strategies to operate without LiDAR if venue conditions prevent use.

written by shuji

2026-03-01

Tasks performed

- Lowered Raspberry Pi Camera position to avoid ToF interference and secure space for a display.



Conclusion & next plan

- Verify camera images to ensure field of view is sufficient.

written by shuji

2026-03-08,09

Tasks performed

- Verified display using a test microcontroller for the display board design.

Issues & solutions

Issue	Solution
Display test with Nucleo F303K8 failed, but worked on ESP32.	Likely memory shortage; decided to use STM32F446RE for display verification and board design.

Conclusion & next plan

- Use STM32F446RE for display verification and design a display board accordingly.

written by shuji

2026-04-03

Tasks performed

- Purchased TSD10 and tested with Arduino Mega 2560.

Issue

Issue	Solution
Packet structure does not match the datasheet.	Arduino Mega may lack clock speed; try on STM32F446RE.

Received packets looked like this:

9C 3F 00 80
9C C1 00 7E
9C 81 00 7E
9C 7A 00 85
DC 7A 00 85
9C 3F 00 80
9C 7F 00 C0
9C C2 00 7D
9C 3F 00 80

The header should be 5C. The checksum calculation appears correct.

2026-04-24

Tasks performed

- Refactored STM32 classes.

Conclusion

- Improved error output, fixed type hints, and strengthened communication error tolerance.
- Continue refactoring the next days.

written by shuji

2026-04-28

Tasks performed

- Experiments on slope control.

Conclusion

- Time-based control:
  - Calculated slope horizontal and vertical distances from gyro-obtained angle and baseline speed on level ground.
  - Values were about 20% larger than expected; movement speed differs between level and slope.
- ToF-based control:
  - Measured distance to the floor with rear ToF and derived movement distance; achieved ~5% accuracy.
  - Field-of-view limits make this method difficult for long slopes; measure the first 1–2 tiles with rear ToF while moving, then assume constant speed afterwards.

written by shuji

2026-04-29

Tasks performed

- Field construction: made 3rd-floor walls and two 2nd-floor blocks.

written by shuji

2026-05-04

Tasks performed

- Added support for 3D maps.
- Added z coordinate (layer index) to mazeMap. Layers are assigned sequentially as discovered (0,1,2,...), not necessarily in vertical order.
- Stored per-layer info in layerInfo, including x_offset, y_offset, altitude relative to layer 0.
- When a slope is detected, decide whether moved to an existing layer or a new layer using vertical distance; compute XY offset using horizontal distance and reflect into maps.
- Managed slope transitions by including neighbor cells that point to tiles on different layers; switched mazeAsGraph representation from set() to dict keyed by absolute direction.

Conclusion & next plan

- mazeMap now handles slopes; adjust Dijkstra cost handling for slopes and refine moveTile calculations for horizontal/vertical distances.

written by shuji

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1. https://youtube.com/shorts/yjniMek3LMw ↩
2. https://youtube.com/shorts/pZukrdztOC0 ↩
3. https://www.youtube.com/watch?v=_25MeoID3pc ↩
4. https://youtu.be/cGURSnNEfI8 ↩
5. https://youtu.be/mSuZB_V_7dA ↩
6. https://youtu.be/zmTCaVvwPds ↩
7. https://youtube.com/shorts/y9CP6Ja2mRU ↩
8. https://youtu.be/vVNjoCeSmk0 ↩
9. https://youtube.com/shorts/OL0_sFIBJAA ↩
10. https://youtube.com/shorts/6vsZAsEav1Is ↩
11. https://www.youtube.com/watch?v=YAJcgg2gsr8 ↩
12. https://youtube.com/shorts/fPRcTKm_Zy4 ↩
13. https://youtube.com/shorts/Fdahk38rR18?si=H7itiwNuw8LfTVrO ↩
14. https://youtube.com/shorts/yQDgVs9pxWI ↩
15. https://www.youtube.com/watch?v=gc3BP7pH4bI ↩
16. https://youtu.be/Pr_9n9FL4Lc ↩
17. https://youtu.be/5eBND1GK3mo?si=6sAXkZLTHRTv5KTL ↩
18. https://youtu.be/CT6xOlg6ZW4 ↩
19. https://youtu.be/UyA_Rh-4MZI ↩
20. https://youtube.com/shorts/pdFYbMMHLh4 ↩
21. https://youtu.be/06iScre4Cys ↩